Notes of Optimization

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Contents

Chapter 1

Lecture 1 of 03/03/2026

We will give a quick summary on basic notions of topology, calculus 1 & 2.

Definition 1 (Norm/scalar product). Given $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ the Euclidian norm (or ℓ^2 norm) is defined as

$$\|x\| = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2}.$$

For $x, y \in \mathbb{R}^n$ the *scalar product* is given by

$$\langle x, y \rangle = x \cdot y = x^\top y = \sum_{i=1}^{n} x_i y_i.$$

Moreover, the scalar product induces a norm. Indeed, one can verify that

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Definition 2 (Open ball). Given $x \in \mathbb{R}^n$, $r > 0$ we call the *open ball of radius r centered in* x the set

$$B(x, r) = \{y \in \mathbb{R}^n : \|y - x\| < r\}.$$

Definition 3 (Limit of a sqeuence). Let $\{x_n\}_{n \in \mathbb{N}} \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that x_n converges to $\bar{x}$, and write $x_n \to \bar{x}$ if for every $\varepsilon > 0$ there exists an $N > 0$ such that for every $k > N$, $x_k \in B(\bar{x}, \varepsilon)$, i.e. for k big enough, $\|\bar{x} - x_k\| < \varepsilon$, and we say that

$$\lim_{k \to +\infty} x_k = \bar{x}.$$

Remark 4. The limit in $\mathbb{R}^n$ is the pointwise limit, so $x_k \to \bar{x}$ if and only if $(x_k)_j \to (\bar{x})_j$ for any $1 \leq j \leq n$.

Example 5. Consider the sequence $\{x_k\}_{k \in \mathbb{N}}$ defined as

$$x_k = \left(\frac{1}{k}, \frac{1}{k} \sin(k) \right) \to (0, 0)$$

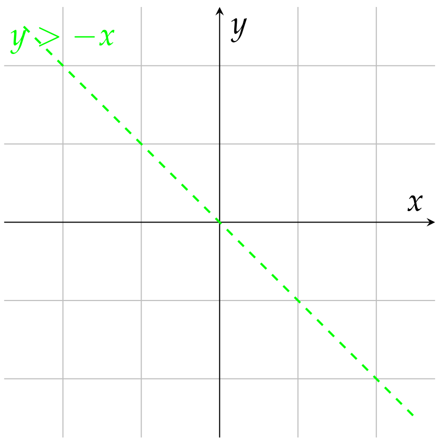

Figure 1.1: Ball of radius r centered in x on $\mathbb{R}^2$.

as $k \to +\infty$.

Definition 6 (Open/closed set)**.** Let $A \subseteq \mathbb{R}^n$. We say that A is *open* if for every $x \in A$ there exists an $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We say that A is *closed* if $\mathbb{R}^n \setminus A$ is open.

Example 7. Consider the set

$$A = \left\{ (x_1, x_2) \in \mathbb{R}^2 : |x_1| < 1, |x_2| < 1 \right\}.$$

We can easily verify that A is open. Observe that a set can be neither open nor closed. The sets $\mathbb{R}^n$ and $\emptyset$ are both open and closed at the same time.

Definition 8 (Neighbour)**.** We say that, for a given $x \in \mathbb{R}^n$, a subset $A \subseteq \mathbb{R}^n$ is a *neighbour* of x if A is an open set containing x.

Definition 9 (Interior, exterior, boundary, closure)**.** Let $A \subseteq \mathbb{R}^n$, $x \in \mathbb{R}^n$. We say that x is:

- an *interior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$. We denote the set of all interior points of A with A°;
- A *boundary point* for A if $\forall \varepsilon > 0$, $B(x, \varepsilon) \cap A \neq \emptyset$ and $B(x, \varepsilon) \cap A^C \neq \emptyset$. We denote the set of all the boundary points with ∂A;
- an *exterior point* for A if $\exists \varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A^C$.

We denote the *closure* of A as $\bar{A} = A \cup \partial A$.

Proposition 10. *Let $\{A_k\}_{k \in \mathbb{N}}$ a sequence of open sets. Then:*

i) *the countable union of open sets is open, i.e.*

$$\bigcup_{k \in \mathbb{N}} A_k \text{ is open;}$$

ii) *the* finite intersection *of open sets is open, i.e. for* $N \in \mathbb{N}$,

$$\bigcap_{i=1}^{N} A_i \text{ is open.}$$

Proof. We prove *i*) and *ii*) separately.

i): set $E = \bigcup_{k=1}^{+\infty} A_k$. Given $x \in A$ then there exits $m > 0$ such that $x \in A_m$. Since A_m is open, there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A_m \subseteq E$, so we found an open ball containing x fully contained in E. Since the choice of x is arbitrary, this must hold for every $x \in E$, hence E is open.

ii): set $E = \bigcap_{k=1}^{N} A_k$. Let $x \in E$, then $x \in A_k$ for every $1 \leq k \leq N$, hence there exists a sequence $\varepsilon_k > 0$ such that $B(x, \varepsilon_k) \subseteq A_k$ for every $k < N$. Consider $\bar{\varepsilon} = \frac{1}{2} \min_k \varepsilon_k > 0$. Then $B(x, \bar{\varepsilon}) \subseteq B(x, \bar{\varepsilon})$ for every k, hence $B(x, \bar{\varepsilon}) \subseteq E$. Again, since the choice of x is arbitrary, this must hold true for every $x \in E$, hence E is open.

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Proposition 11 (Characterization of closed sets). *Let* $C \subseteq \mathbb{R}^n$. *Then* C *is closed* $\iff$ *for every sequence* $\{x_k\}_{k \in \mathbb{N}} \subseteq C$ *such that* $x_k \to \bar{x}$, *then* $\bar{x} \in C$.

Proof. We prove $\implies$ and $\impliedby$.

" $\implies$ **":** let $C \subseteq \mathbb{R}^n$ be closed. Consider $\{x_k\}_{k \in \mathbb{N}}$ such that $x_k \to \bar{x} \in \mathbb{R}^n$, we wanto to show that $\bar{x} \in C$. Assume, by contraddiction, that $\bar{x} \notin C \implies \bar{x} \in \mathbb{R}^n \setminus C$, which is an open set, so there exists $\varepsilon > 0$ such that $B(\bar{x}, \varepsilon) \subseteq \mathbb{R}^n \setminus C \implies B(\bar{x}, \varepsilon) \cap C = \emptyset$, but $x_k \to \bar{x}$, so, for this ε, there must be an $N > 0$ such that $x_k \in B(\bar{x}, \varepsilon)$ for $k > N$, which is in contraddiction with the fact that the sequence takes values in C.

" $\Longleftarrow$ ": let C be a set. Consider $\{x_k\}_{k\in\mathbb{N}} \subseteq C$ such that $x_k \to \bar{x} \in C$. Assume, by contraddiction, C not closed. Then, for every $\varepsilon > 0$, there exists $z \in \mathbb{R}^n \setminus C$ such that $B(z,\varepsilon) \cap (\mathbb{R}^n \setminus C) = \emptyset$ and $B(z,\varepsilon) \cap C \neq \emptyset$. So we can find a sequence $\{x_k\}_{k\in\mathbb{N}}$ with values in C such that $x_k \to z$ by fixing a sequence $\{\varepsilon_k\}_{k\in\mathbb{N}}$ such that $\varepsilon_k \searrow 0$, so $x_k \in B(z,\varepsilon_k) \cap C$, $x_k \to z$, so, by assumption $z \in C$, which is in contraddiction with $z \in \mathbb{R}^n \setminus C$.

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Definition 12 (Bounded/finite set)**.** A set $A \subseteq \mathbb{R}^n$ is *bounded* if there exists $R > 0$ such that $A \subseteq B(0,R)$.

The set A is *finite* if $A = \{x_1, \dots, x_m\} \subseteq \mathbb{R}^n$.

Definition 13 (Continuity)**.** Let $f : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $\bar{x} \in E$. We say that f is *continous* in $\bar{x}$ if for every sequence $\{x_k\}_{k\in\mathbb{N}}$ such that $x_k \to \bar{x}$ in $\mathbb{R}^n$, then $f(x_k) \to f(\bar{x})$ in $\mathbb{R}^m$.

Definition 14 (Cluster/isolated point)**.** Let $E \subseteq \mathbb{R}^n$, $\bar{x} \in \mathbb{R}^n$. We say that $\bar{x}$ is a *cluser point* for E if there exists a sequence $\{x_k\}_{k\in\mathbb{N}}$ such that, for every $\bar{x} \neq x_k \to \bar{x}$. We denote tha set of all cluster points with $\mathrm{Acc}(E)$.

A point $\bar{x}$ is *isolated* in E if $x \in E \setminus \mathrm{Acc}(E)$.

Definition 15 (Limit)**.** Let $f : E \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $\bar{x} \in \mathrm{Acc}(E)$. Fixed $\ell \in \mathbb{R}^m$, we say that

$$\lim_{x\to\bar{x}} f(x) = \ell$$

if for every sequence $\{x_k\}_{k\in\mathbb{N}}$, $x_k \to \bar{x}$, $x_k \neq \bar{x}$ for every k, we have $f(x_k) \to \ell$. Notice that we allow $\bar{x}$ to not necessarely be in E.

Example 16. Consider the function $f : \{0\} \cup [1,2] \subset \mathbb{R} \to \mathbb{R}$, $x \mapsto x$. Then it has no meaning to speak about $\lim_{x\to 0} f(x)$, as $0 \notin \mathrm{Acc}(\{0\} \cup [1,2])$, while f is continous at 0, since the only sequence converging to 0 in $\{0\} \cup [1,2]$ is the sequence which is identically 0.

Remark 17. If $f : \mathbb{R}^n \to \mathbb{R}$ is continous, then:

- $\{x : f(x) < c\}$ is *open* $\forall c \in \mathbb{R}$;
- $\{x : f(x) \leq c\}$ is *closed* $\forall c \in \mathbb{R}$;
- $\{x : f(x) = c\}$ is *closed* $\forall c \in \mathbb{R}$.

Chapter 2

Lecture 2 of 04/03/2026

Our main focus from now on is to solve the problem

$$\inf_{x\in A} f(x)$$

for some $A \subseteq \mathbb{R}^n$ and some $f : \mathbb{R}^n \to \mathbb{R}$. We will restrict ourselfs only to certain functions and sets.

Definition 1 (Infimum/supremum)**.** Let $E \subseteq \mathbb{R}$. We denote with $\inf E =$ largest $z \in \mathbb{R}$ such that $z \leq x$ for every $x \in E$. By convention we set $\inf \emptyset = +\infty$.
Similarly, we denote with $\sup E$ the smallest $z \in \mathbb{R}$ such that $z \geq x$ for every $x \in E$, with the convention $\sup \emptyset = -\infty$. More formally,

$$\begin{aligned} z = \inf E \quad & \text{if } z \leq x \, \forall x \in E \text{ and } \forall \varepsilon > 0 \, \exists y \in E \text{ s.t. } y \leq z + \varepsilon \\ z = \sup E \quad & \text{if } z \geq x \, \forall x \in E \text{ and } \forall \varepsilon > 0 \, \exists y \in E \text{ s.t. } y \geq z - \varepsilon \end{aligned}$$

Remark 2. The inf and sup aren't necessarely inside the set. If they are, we speak of min and max.

Axiom 1 — *Completeness of $\mathbb{R}$*

Every bounded sequence $\{x_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}$ admits a supremum in $\mathbb{R}$.

Theorem 3 (Bolzano-Weierstrass)**.** *Any bounded sequence $\{x_n\}_{n\in\mathbb{N}} \subseteq \mathbb{R}$ admits a cluster point, i.e. there exists $\bar{x} \in \mathbb{R}$ and a subsequence $\left\{x_{n_j}\right\}_{j\in\mathbb{N}}$ such that $x_{n_j} \to \bar{x}$ as $j \to +\infty$.*

Proof. Since $\{x_n\}_n$ is bounded, there exists $R > 0$ such that $x_n \in [-R, R]$ for every $n \in \mathbb{N}$. Set $I_1 = [-R, R]$, select x_{n_1} a value in I_1. Divide I_1 in half and select the one subinterval containing inifitely many x_ns, call it $I_2 = [a_2, b_2]$ (here $a_2 = -R, b_2 = 0$ or $a_2 = 0, b_2 = R$). We have $I_2 \subset I_1$, $|I_2| = |I_1|/2$, and choose x_{n_2} a value in I_2. Repeat this process to cut and select. We end up, at the k-th step, with a sequence of subintervals $I_1, I_2, \ldots, I_k$ and 3 subsequences: $x_{n_1}, x_{n_2}, \ldots, x_{n_k}$

the sequence of selected values, $a_1, a_2, \dots, a_k$ the sequence of left boundaries and $b_1, b_2, \dots, b_k$ the sequence of right boundaries.

We observe that $a_{j+1} \geq a_j$ and $b_{j+1} \leq b_j$ for all $1 \leq j \leq k$. Moreover, $a_j \leq R$ and $b_j \geq -R$ for all j, so, by the axiom of completeness of $\mathbb{R}$,

- there exists $\bar{a} \in \mathbb{R}$ such that $a_k \to \bar{a} = \sup_k a_k$
- there exists $\bar{b} \in \mathbb{R}$ such that $b_k \to \bar{b} = \inf_k b_k$

We observe that $\bar{a} = \bar{b}$. Indeed, $\left|\bar{b} - \bar{a}\right| = \left|\bar{b} - b_k + b_k - a_k + a_k - \bar{a}\right| \leq \left|\bar{b} - b_k\right| + |\bar{a} - a_k| + |b_k - a_k|$. The terms $\left|\bar{b} - b_k\right|$ and $|\bar{a} - a_k|$ goes to 0 as $k \to +\infty$ for what we observed before. Moreover, $|b_k - a_k| = 2R/2^{k-1} \to 0$ as $k \to +\infty$.

Set $\bar{x} = \bar{a} (= \bar{b})$. Then the subsequence $\left\{x_{n_j}\right\}_j$ is converging to $\bar{x}$: indeed $\left|x_{n_j} - \bar{x}\right| = \left|x_{n_j} - a_j + a_j - \bar{x}\right| \leq \left|a_{n_j} - a_j\right| + \left|a_j - \bar{x}\right|$.

Since $a_j \to \bar{x} = \bar{a}$, the term $\left|a_j - \bar{x}\right| \to 0$. Notice that, by construction, $x_{n_j} \in I_j$, so $\left|x_{n_j} - a_j\right| \leq \left|I_j\right| \to 0$.

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Definition 4. Let $f : \mathbb{R}^n \to \mathbb{R}$. We say that $\bar{x} \in \mathbb{R}^n$ is a *local maximum* for f if there exists a neighbourhood U of $\bar{x}$ such that $f(x) \leq f(\bar{x})$ for every $x \in U$. A *local strict maximum* has the strict inequality.

Similarly, $\bar{x}$ is *local minumum* for f if $\exists U$ neighbourhood of $\bar{x}$ such that $f(x) \geq f(\bar{x})$ for all $x \in U$.

If there exists $\bar{x}$ such that $f(x) = \inf_{x \in A} f(x)$, the $\bar{x}$ is a *global minimum* over A.

Definition 5. We denote with $\arg_{x \in A} \min f(x)$ the set of minumum points for f over A, i.e

$$\arg_{x \in A} \min f(x) = \left\{\bar{x} \in \mathbb{R}^n : f(\bar{x}) = \inf_{x \in A} f(x)\right\}.$$

Theorem 6 (Weierstrass). *Let $A \subseteq \mathbb{R}^n$ be a compact set, $f : A \to \mathbb{R}$ a continous function. The there exists x_m, $x_M \in A$ such that*

$$f(x_m) = \min_{x \in A} f(x)$$
$$f(x_M) = \max_{x \in A} f(x).$$

Proof. We prove the statement for the minimum point.

Let $\ell = \inf_{x \in A} f(x)$. We can always find a sequence $\{x_n\}_n \subseteq A$ such that $f(x_n) \to \ell$ by the continuity of f. Since A is bounded, $\{x_n\}_n$ is bounded as well. By Bolzano-Weierstrass, there exists a subsequence $\{x_{n_k}\}_k$ converging to $\bar{x} \in A$ by closerness of A. By continuity, in particular f is continous at $\bar{x}$,

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Remark 7. For $f : \mathbb{R}^n \to \mathbb{R}$, we define the *gradient* of f as

$$\nabla f(c) = \left(\frac{\partial f}{\partial x_1}(c), \dots, \frac{\partial f}{\partial x_n}(c) \right)$$

the vector of partial derivatives, defined as

$$\frac{\partial f}{\partial x_i}(c) = \lim_{h \to 0} \frac{f(c + he_i) - f(c)}{h}$$

whenever this limit exists finite, and where e_i is the i-th vector of the canonical base of $\mathbb{R}^n$. Furthermore, we say that f is *differentiable* at c if

$$f(x) = f(c) + \langle \nabla f(c), x - c \rangle + \mathcal{O}(\|x - c\|^2)$$

for $x \to c$.

We now set ouf focus on convex analysis.

Definition 8 (Convex set)**.** Let $C \subseteq \mathbb{R}^n$. We say that C is *convex* if, for any $x, y \in C$, and for any $\alpha \in [0,1]$, $\alpha x + (1 - \alpha) y \in C$.

Proposition 9. *a)* $\bigcap_{i \in I} C_i$ *is convex if* C_i *is convex for any* $i \in I$ *finite or countable;*

b) the algebraic sum $C_1 + C_2 = \{x + y : x \in C_1,\ y \in C_2\}$ *is convex if* C_1 *and* C_2 *are convex;*

c) the set $\lambda C = \{\lambda x : x \in C\}$ *is convex if* C *is convex for any* $\lambda \in \mathbb{R}$*;*

d) $\overline{C}, C^\circ$ *are convex if* C *is convex;*

e) if a function $f : C \to \mathbb{R}$ *is affine, then the set* $f(C) = \{y : y = f(x),\ x \in C\}$ *is convex if* C *is convex.*

Proof. We prove all the propositions separately.

a): suppose C_i convex and let $C = \bigcap_{i \in I} C_i$. Let $x, y \in C$, $\alpha \in [0,1]$. Since $x, y \in C$, in particular $x, y \in C_i$ for all $i \in I$. But then also $\alpha x + (1-\alpha)y \in C_i$ for all $i \in I$, hence C is convex.

b): let $x = x_1 + x_2 \in C_1 + C_2$, $x_1 \in C_1$ and $x_2 \in C_2$, take $y \in C_1 + C_2$ of the same form.

$$\alpha x + (1 - \alpha y) = \underbrace{\alpha x_1 + (1-\alpha)y_1}_{\in C_1} + \underbrace{\alpha x_2 + (1-\alpha)y_2}_{\in C_2} \in C.$$

c): trivial.

d): consider $\{x_k\}_k \subseteq C$ and $\{y_k\}_k \subseteq C$ such that $x_k \to x$ and $y_k \to y$. Then the sequence $\{\alpha x_k + (1-\alpha)y_k\}_k \subseteq C$. But then also $\alpha x + (1-\alpha)y \in C$ as limit of sequences valued in C.

Let now $x, y \in C^\circ$. Then there exists $r > 0$ such that $B(x,r) \subseteq C$ and $B(y,r) \subseteq C$. Consider $\alpha x + (1-\alpha)y$. Our claim is: $B(\alpha x + (1-\alpha)y, r) \subseteq C$. Indeed, any $z \in B(\alpha x + (1-\alpha)y, r)$ is of the form $z = \alpha x + (1-\alpha)y + w$ with $\|w\| < r$, so $\alpha(x+w) + (1-\alpha)(y+w) \in C$ by convexity of C, hence the claim.

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Chapter 3

Lecture 3 of 10/03/2026

Example 1 (Convex sets)**.** The following sets are convex in $\mathbb{R}^n$:

- **hyperplanes**: $\left\{x \in \mathbb{R}^n : a^\top x = b\right\}$ for $a \in \mathbb{R}^n$, $b \in \mathbb{R}$;
- **halfspaces**: $\left\{x \in \mathbb{R}^n : a^\top x \leq b\right\}$ for $a \in \mathbb{R}^n$, $b \in \mathbb{R}$;
- **polyhedra**: $\left\{x \in \mathbb{R}^n : a_i^\top x \leq b_i\right\}$ for $a_i \in \mathbb{R}^n$, $b_i \in \mathbb{R}$ for $i = 1, \ldots, m$ (they are intersecttion of half spaces).

Half cones are convex as well.

Definition 2 (Convex/concave function)**.** Let $C \subseteq \mathbb{R}^n$ be convex. Consider $f : C \to \mathbb{R}$. We say that f is *convex* if the following holds:

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y) \quad \forall x, y \in C,\ \forall \alpha \in [0,1].$$

We say that f is *concave* if $-f$ is convex.

Example 3. Affine functions are convex. Indeed, let $f(x) = a^\top x + b$, $a \in \mathbb{R}^n$. Indeed,

$$\begin{aligned} f(\alpha x + (1-\alpha)y) &= \langle a, \alpha x + (1-\alpha)y \rangle + b(1 - \alpha + \alpha) \\ &= \alpha \langle a, x \rangle + \alpha b + (1-\alpha)\langle a, y \rangle + (1-\alpha) b \\ &= \alpha f(x) + (1-\alpha) f(y). \end{aligned}$$

Example 4. The norm in $\mathbb{R}^n$ is convex, as an easy consequence of triangular inequality.

We can observe that f convex implies that the set $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ is convex for all $\gamma \in \mathbb{R}$. Indeed, for every $x, y \in V_\gamma$ we have $f(x) \leq \gamma$ and $f(y) \leq \gamma$. We see that $f(\alpha x + (1-\alpha)y) \in V - \gamma$. Indeed, by convexity of f we have $f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y) = \alpha\gamma + (1-\alpha)\gamma = \gamma$.

In general, the converse is not true, i.e. convex level sets *does not imply* f convex.

From now on, $f : \mathbb{R}^n \to \overline{\mathbb{R}}$, where $\overline{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$.

Definition 5 (Epigraph). Let $f : \mathbb{R}^n \to \overline{\mathbb{R}}$. We define the set

$$\operatorname{epi}(f) = \left\{(x, w) \in \mathbb{R}^{n+1} : x \in X,\, w \in \mathbb{R},\, f(x) \leq w\right\}.$$

An intuitive definition could be "*the set of points above the graph of f*".

Definition 6 (Effective domain). Given $f : X \subseteq \mathbb{R}^n \to \overline{\mathbb{R}}$ we define the *effective domain* of f as the set

$$\operatorname{dom}(f) = \{x \in X : f(x) < +\infty\} = \{x \in X : \exists\, w \in \mathbb{R} \text{ s.t. } (x, w) \in \operatorname{epi}(f)\}.$$

Definition 7 (Proper function). A function $f : X \subseteq \mathbb{R}^n \to \overline{\mathbb{R}}$ is said *proper* if $f(x) < +\infty$ and $f(x) > -\infty$ for all $x \in X$. In other words, $\operatorname{epi}(f)$ does not contain any vertical line.

Definition 8 (Extension of convex functions). Let $C \subseteq \mathbb{R}^n$ be a convex set. A funciton $f : C \to \overline{\mathbb{R}}$ is *convex* if $\operatorname{epi}(f)$ is convex in $\mathbb{R}^{n+1}$.

Remark 9. If $f : C \to \mathbb{R}$ the definitions coincides.

Proof. Assume $\operatorname{epi}(f)$ convex in $\mathbb{R}^{n+1}$, then the points $(x, f(x)), (y, f(y)) \in \operatorname{epi}(f)$, but also $\alpha(x, f(x)) + (1-\alpha)(y, f(y)) \in \operatorname{epi}(f)$. This implies $(\alpha x + (1-\alpha)y, \alpha f(x) + (1-\alpha)f(y)) \in \operatorname{epi}(f) \implies f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$. ∎

Definition 10 (Convex indicator function). Given $A \subseteq \mathbb{R}^n$, we define the *convex indicator function* of the set A as

$$\delta_A(x) = \begin{cases} 0 & \text{if } x \in A \\ +\infty & \text{otherwise} \end{cases}$$

This is coherent with the infimum operation, as

$$\inf_{x \in A} f(x) = \inf_{x \in \mathbb{R}^n} f(x) + \delta_A(x).$$

Definition 11 (Liminf). Let $f : x \subseteq \mathbb{R}^n \to \mathbb{R}$, $c \in \operatorname{Acc}(X)$. We say that

$$\ell = \liminf_{x \to c} f(x)$$

if $\ell = \inf_{x_k} \lim_{k \to +\infty} f(x_k)$ where $\{x_k\}_k$ is a sequence in X, $\{x_k\}_k \to c$, $x_k \neq c$ for all k and $\lim_{k \to \infty} f(x_k)$ exists finite.

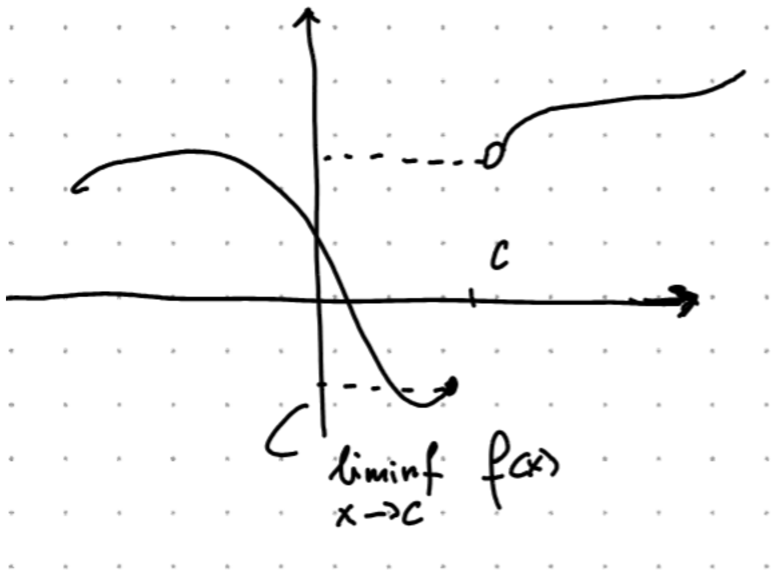

Figure 3.1: The liminf of f at $x = 1$ is 1, even if $f(1) = 0.5$.

Definition 12 (Lower semicontinuity). Let $f : X \subseteq \mathbb{R}^n \to \mathbb{R}$. We say that f is *lower semicontinuous* (lsc for short) at $c \in X$ if

$$f(c) \leq \liminf_{x \to c} f(x).$$

Theorem 13 (Weierstrass). *The Weierstrass theorem still allows a minimum point for a lsc funciton.*

Proposition 14. *Let $f : \mathbb{R}^n \to \overline{\mathbb{R}}$. The following are equivalent:*

$i)$ The level sets $V_\gamma = \{x \in \mathbb{R}^n : f(x) \leq \gamma\}$ are closed for all $\gamma \in \mathbb{R}$;

$ii)$ f is lsc;

$iii)$ $\operatorname{epi}(f)$ is closed in $\mathbb{R}^{n+1}$.

Proof. We prove $i) \implies ii) \implies iii) \implies i)$.

$i) \implies ii)$**:** suppose V_γ closed for all γ. Assume by contraddiction f not lsc, so there exists $c \in \mathbb{R}^n$ s.t. $f(c) > \liminf_{x \to c} f(x)$. This implies that there exists a sequence $\{x_k\}_k$, $x_k \to c$ and a $\gamma \in \mathbb{R}$ s.t. $f(c) > \gamma \geq \lim_{k \to +\infty} \inf f(x_k)$. For k big enough , we have $f(x_k) \leq \gamma$, so $x_k \in V_\gamma$. By closerness of V_γ, it must be $c \in V_\gamma \implies f(c) \leq \gamma$, which is in contraddiction with $f(c) > \gamma$.

ii) $\implies$ *iii)*: assume f lsc, we want to show that any sequence $\{(x_k, w_k)\}_k \subseteq \text{epi}\,(f)$ converging to $(\bar{x}, \bar{w})$ must satisfy $(x, w) \in \text{epi}\,(f)$. Assime $x_k \to \bar{x}$, $w_k \to \bar{w}$. Then $f(x_k) \leq w_k$ for all k. Since $x_k \to \bar{x}$ and f is lsc, it must be that $f(\bar{x}) \leq \liminf_{k \to +\infty} f(x_k) \leq \liminf_{k \to +\infty} w_k = \bar{w}$, hence $(\bar{x}, \bar{w}) \in \text{epi}\,(f)$.

iii $\implies$ *i)*: assume epi (f) closed. Let $\gamma \in \mathbb{R}$, $V_\gamma = \{x : f(x) \leq \gamma\}$. Consider the sequence $\{x_k\}_k \subseteq V_\gamma$ s.t $x_k \to \bar{x}$. We want to show $\bar{x} \in V_\gamma$. The points $(x_k, \gamma) \in \text{epi}\,(f)$ for all k, so, by closerness of epi (f), $(\bar{x}, \gamma) \in \text{epi}\,(f)$. This impies $f(\bar{x}) \leq \gamma$, hence $\bar{x} \in V_\gamma$.

■

Proposition 15. *Let $f : \mathbb{R}^m \to (-\infty, +infty]$, $A \in \mathbb{R}^{m \times n}$, $F : \mathbb{R}^n \to (-\infty, +\infty]$, $x \mapsto f(Ax)$. If f is convex, then F is convex. If f is closed, then F is closed.*

Proof. Assume f convex. Then

$$F(\alpha x + (1 - \alpha)y) = f(\alpha Ax + (1 - \alpha)Ay) \leq \alpha f(Ax) + (1 - \alpha)f(Ay).$$

Assume f closed, so epi (f) is closed and f is lsc. Take a sequence $x_k \to \bar{x}$. Then $\liminf_k F(x_k) = \liminf_k f(Ax_k)$ Let $y_k = Ax_k$, so $y_k \to A\bar{x}$ by continuity of the linear map Ax. This implies $\liminf_k f(Ax_k) \geq f(A\bar{x}) = F(\bar{x})$.

■

Proposition 16. *Let $\{f_i\}_{i=1}^m$ be a family of convex and closed function s.t. $f_i : \mathbb{R}^n \to (-\infty, +\infty]$, and $\gamma_1, \ldots, \gamma_m > 0$. Then the function defined by $F(x) = \sum_{i=1}^m \gamma_i f_i(x)$ is closed and convex.*

Proposition 17. *Let $f_i : \mathbb{R}^n \to (-\infty, +\infty]$ for $i \in I$ any countable index set. Let $f(x) = \sup_i f_i(x)$. If f_i is closed convex for all $i \in I$ then f is closed convex.*

Proof. Let $(x, w) \in \text{epi}\,(f)$, then $f(x) \leq w$, so $f_i(x) \leq w$ for all i. This implies $(x, w) \in \text{epi}\,(f_i)$ for all i, hence $\text{epi}\,(f) = \bigcap_{i \in I} \text{epi}\,(f_i)$, which is convex as countable intersection of convex sets. The same argument holds for closerness.

■

Chapter 4

Lecture 4 of 11/03/2026

We ask ourselfs what happens if f is convex and we add some regularity hypotesis. We first remark that if f is differentiable at c, near this point, the function looks like an hyperplane. The intuition is: function must be always above its tangent hyperplane.

Proposition 1. *Let $C \subseteq \mathbb{R}^n$ be convex, $A \subseteq \mathbb{R}^n$ open s.t. $C \subseteq A$. Let $f : A \to \mathbb{R}$ convex and differentiable. Then*

i) f is convex on $C \iff f(z) \geq f(x) + \langle \nabla f(x), z - x\rangle$ for all $x, z \in C$

ii) f is strictly convex over C $\iff f(z) > f(x) + \langle \nabla f(x), z - x\rangle$ for all $x, z \in C$ s.t. $x \neq z$

Proof. Let f be convex on C. Let $x, z \in C$. Observe that

$$\lim_{h\to 0^+} \frac{f(x + h(z - x)) - f(x)}{h} = \frac{\partial}{\partial h} f(x + h(z - x))|_{h=0} = \nabla f(x) \cdot (z - x).$$

Then

$$\begin{aligned}
f(x) + \langle \nabla f(x), z - x\rangle &= f(x) + \lim_{h\to 0^+} \frac{f(x + h(z - x)) - f(x)}{h} \\
&= \lim_{h\to 0} \frac{hf(x) - f(x) + f(hz + (1 - h)x)}{h} \\
&\leq \lim_{h\to 0} \frac{-(1 - h)f(x) + hf(z) - (1 - h)f(x)}{x} \\
&= \lim_{h\to 0} f(z) = f(z).
\end{aligned}$$

Let now $\alpha \in [0, 1]$, $x, y \in C$, $z = \alpha x + (1 - \alpha)y$.

$$\begin{aligned}
f(x) &\geq f(z) + \langle \nabla f(z), x - z\rangle \cdot \alpha \\
f(y) &\geq f(z) + \langle \nabla f(z), y - z\rangle \cdot (1 - \alpha)
\end{aligned}$$

Take the sum of the above quantities, then

$$\alpha f(x) + (1-\alpha)f(y) \geq \alpha f(z) + (1-\alpha)f(z) + \alpha\langle \nabla f(z), x-z\rangle + (1-\alpha)\langle \nabla f(z), y-z\rangle$$

hence, by linearity of the scalar product,

$$\alpha f(x) + (1-\alpha)f(y) \geq f(z) + \langle \nabla f(z), \alpha x + (1-\alpha)y - z\rangle.$$

Hence, f is convex.

For the strict convexity, we repeat the same process with the strict inequality. ■

If at some point x^* we have $\langle \nabla f(x^*), z - x^*\rangle \geq 0$ for all z, then x^* is a minimum point. This is also a necesary condition: assume x^* minumum point for f over C and, by contraddiction, thath there exists $z \in C$ s.t. $\langle \nabla f(x^*), z - x^*\rangle < 0$, but then

$$\frac{\partial}{\partial h} f(x^* + h(z - x^*))|_{h=0} = \langle \nabla f(x^*), z - x^*\rangle < 0$$

so f decreases in the direction $z - x^*$, which contrdaddicts the fact that x^* is a minimum.

We summarize this fact in the following proposition, which proof what we just did.

Proposition 2. *Let $C \subseteq \mathbb{R}^n$ be convex, $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable on a convex open set $A \subseteq C$. Then a point $x^* \in C$ is a minimum for f if and only if*

$$\langle \nabla f(x^*), z - x^*\rangle \geq 0 \quad \forall z \in C.$$

Theorem 3 (Projection theorem)**.** *Let C be a nonempty convex closed sunset of $\mathbb{R}^n$, $z \in \mathbb{R}^n$. Then there exists a unique vector $x^* \in C$ (the projection of z on C) that minimizes the function $f(x) = \|x - z\|$ over C.*

Furthermore, the point x^* is the projection of z onto C if and only if

$$\langle z - x^*, x - x^*\rangle \leq 0 \quad \forall x \in C.$$

Proof. Optimizing the function $f(x) = \|z - x\|$ is the same as optimizing the funciton $f(x) = \frac{1}{2}\|z - x\|^2$. Then f is now differentiable, and $\nabla f(x) = z - x$.

$$\begin{aligned} x^* \text{ is minimum} &\iff \langle \nabla f(x^*), x - x^*\rangle \geq 0 \quad \forall x \in C \\ &\iff \langle -x^* + z, x - x^*\rangle \leq 0. \end{aligned}$$

Does such an x^* exists? We observe that

$$\min_{x\in C} f(x) = \min_{x\in C\cap\{y:\|y-z\|\leq\|z-w\|\},\, w\in C} f(x).$$

We notice that we are optimizing a continous function on a closed set, as we are optimizing on $C \cap \overline{B(z, \|z-w\|)}$, so, by Weierstrass there exists a minimum. This is also unique, indeed, let x_1^*, x_2^* be two minima, then

$$\langle z - x_1^*, x_2^* - x^*1\rangle \leq 0 \quad \forall z \in C$$
$$\langle z - x_2^*, x_2^* - x^*1\rangle \leq 0 \quad \forall z \in C$$

Taking the sum ad using the linearity of the scalar product, we obtain

$$\langle x_1^*, x_2^*, z - x_2^* - z + x_1^*\rangle \leq 0 \implies \|x_1^* - x_2^*\| \leq 0 \implies x_1^* = x_2^*.$$

■

Convex hulls

The idea is ton"convexify" sets that are not convex. Let X be a set. The *convex hull* of X conv $(()\, X)$ = intersection of all convex sets containing X. A convex combination of elements of X is a vector in the form $\sum_{i=1}^m \alpha_i x_i$, where x_i are points in X, $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$.

Remark 4. If X is convex, any convex combination of elements of X belongs to X.

Remark 5.

$$\operatorname{conv}(X) = \left\{ z = \sum_{i=1}^m \alpha_i x_i : \alpha_i > 0,\ x_i \in X,\ \sum_i \alpha_i = 1,\ \forall m > 0 \right\}.$$

Remark 6. If $A : \mathbb{R}^n \to \mathbb{R}^n$ is a linear tranformation, then $\operatorname{conv}(Ax) = A \operatorname{conv}(X)$.

Remark 7. $\operatorname{conv}\left(\sum_i X_i\right) = \sum_i \operatorname{conv}(X_i)$.

Definition 8 (Affine hull)**.** Let $X \subseteq \mathbb{R}^n$. The *affine hull* of X, denoted with $\operatorname{aff}(X)$ is the intersection of all affine sets containing X.

Recall that M is an affine set if M is in the form $M = \bar{x} + S$, where S is a linear subspace of $\mathbb{R}^n$.

Notice that $\operatorname{conv}(X) \subseteq \operatorname{aff}(X)$, and intersection of affine spaces is still an affine space, so $\operatorname{aff}(X)$ is an affine space, so $affX = \bar{x} + S$ for sone $\bar{x} \in \mathbb{R}^n$ and S subspace. We define the dimension of $\operatorname{aff}(X)$ as the dimension of S. S is also called the parallel subspace of the affine space.

Remark 9. $\operatorname{aff}(X) = \operatorname{aff}(\operatorname{conv}(X)) = \operatorname{aff}(\operatorname{cl}(X))$ where $\operatorname{cl}(X)$ denotes the closure of X.

Definition 10 (Cone generated by X)**.** Let $X \subseteq \mathbb{R}^n$. We define the *cone generated by* X the set

$$\operatorname{cone}(X) = \left\{ \sum_{i=1}^{m} \alpha_i x_i : \alpha_i \geq 0,\ x_i \in X,\ m = 0 \right\}.$$

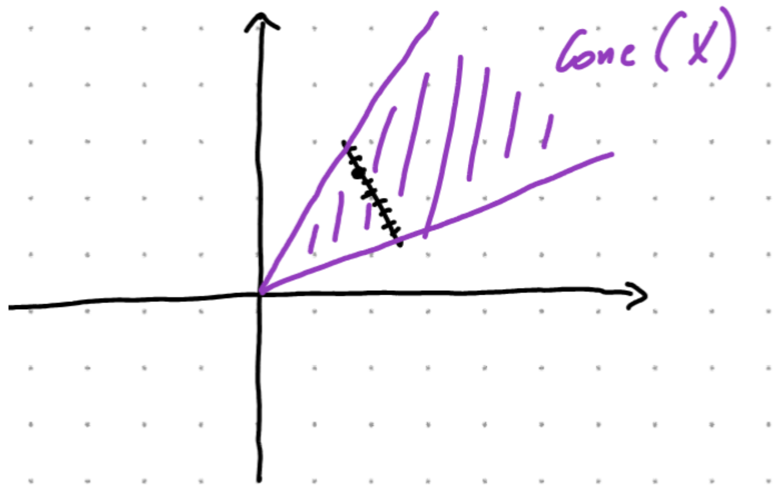

Figure 4.1: X (black) and cone generated by X (purple)

Chapter 5

Lecture 5 of 18/03/2026

Theorem 1 (Charateodory). *Let $X \subseteq \mathbb{R}^n$ be nonempty. Then*

a) *every nonzero vector from* $\operatorname{cone}(X)$ *can be represented as a positive combination of linearly independent vectors in X;*

b) *every vector fron* $\operatorname{conv}(X)$ *can be represented as a convex combination of no more then $n+1$ linearly independent vectors in X.*